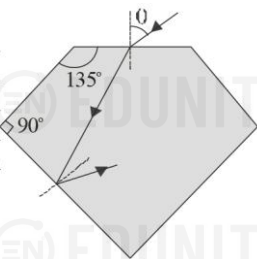
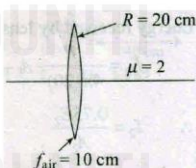


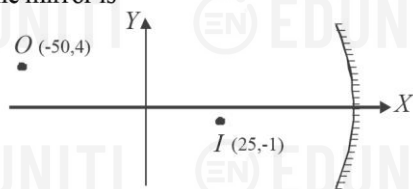
1. A ray of light enters a diamond ($\mu = 2$) from air and is being internally reflected near the bottom as shown in the figure. Find maximum value of angle θ possible.



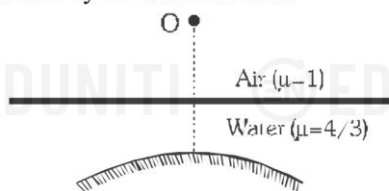
2. Focal length of a thin lens in air is 10 cm. Now, medium on one side of the lens is replaced by a medium of refractive index 2. The radius of curvature of the surface of lens, in contact with the medium, is 20 cm. Find the new focal length of the lens when the object is placed in air on the other side.



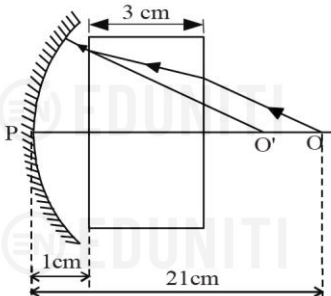
3. A plane mirror is placed 22.5 cm in front of a concave mirror of focal length 10 cm. Find, where an object can be placed between the two mirrors, so that the first image in both mirrors coincide.
4. A concave mirror forms an image I corresponding to a point object O . Find the equation of the circle intercepted by the xy plane on the mirror is



5. A point object lies 30 cm above water on the axis of a convex mirror of focal length 40 cm lying 20 cm below water surface. Find the location of final image formed after refraction by water surface followed by reflection from mirror and again refraction by the water surface.

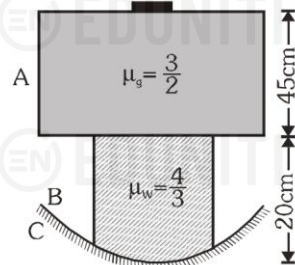


6. An object is placed 21 cm in front of a concave mirror of radius of curvature 10 cm. A glass plate of thickness 3 cm and refractive index 1.5 is then placed close to the mirror in the space between the object and the mirror. Find the position of the final image formed. (You may take the distances of the nearer surface of the plate from the mirror to be 1 cm).

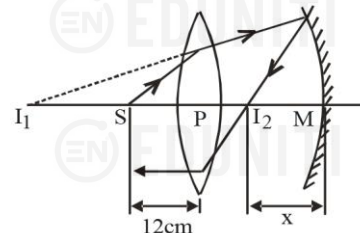


7. A point object is placed 33 cm from a convex mirror of radius of curvature 40 cm. A glass plate of thickness 6 cm and refractive index 2.0 is placed between the object and mirror, close to the mirror. Find the distance of final image from the object.

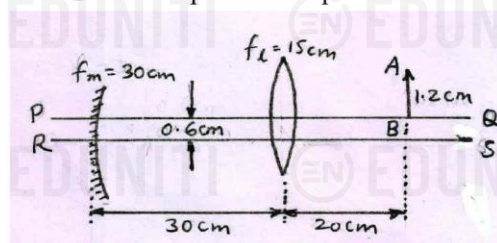
8. A fly F is sitting on a glass slab A , 45 cm thick and of refractive index $3/2$. The slab covers the top of a container B containing water (refractive index $4/3$) upto a height of 20 cm. The bottom of container is closed by a concave mirror C of radius of curvature 40 cm. Locate the final image formed by all refractions and reflection assuming paraxial rays.



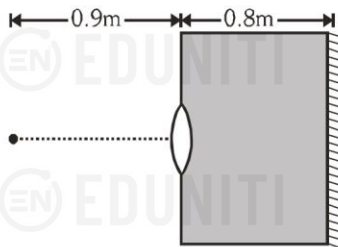
9. A converging lens of focal length 15 cm and a converging mirror of focal length 20 cm are placed with their principal axes coinciding. A point source S is placed on the principal axis at a distance of 12 cm from the lens as shown in figure. It is found that the final beam comes out parallel to the principal axis. Find the separation between the mirror and the lens.



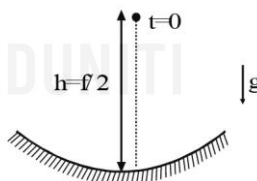
10. A convex lens of focal length 15 cm and a concave mirror of focal length 30 cm are kept with their optic axes PQ and RS parallel but separated in vertical direction by 0.6 cm as shown in fig. The distance between the lens and mirror is 30 cm. An upright object AB of height 1.2 cm is placed on the optic axis PQ of the lens at a distance of 20 cm from the lens. If $A''B''$ is the image after refraction from the lens and reflection from the mirror, find its distance from the pole of the mirror and obtain its magnification. Also, locate positions of A'' and B'' with respect to the optic axis RS .



11. A thin equiconvex lens of glass of refractive index $\mu = 3/2$ and of focal length 0.3 m in air is sealed into an opening at one end of a tank filled with water $\mu = 4/3$. On the opposite side of the lens, a mirror is placed inside the tank on the tank wall perpendicular to the lens axis, as shown in figure. The separation between the lens and the mirror is 0.8 m. A small object is placed outside the tank in front of the lens at a distance of 0.9 m from the lens on its axis. Find the position (relative to the lens) of the image of the object formed by the system.



12. A particle is dropped along the axis from a height $h = f/2$ on a concave mirror of focal length f as shown in figure. Find the maximum speed of image.

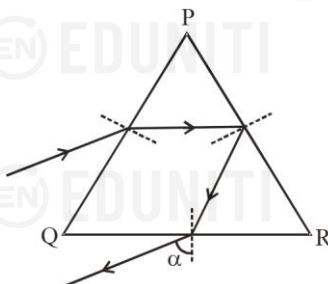


13. A thin equiconvex lens is placed on a horizontal plane mirror and a pin held 20cm above the lens coincides in position with its own image. The space between the lens and mirror is filled with water ($\mu = 4/3$) and then to coincide with its own image as before, the pin has to be raised until its distance from the lens is 27.5cm. Find the radius of curvature of lens

14. A ray of light moving along x -axis is incident on a glass slab at origin at grazing incidence. The refractive index of the material of the slab is given by $\mu = \sqrt{1+y}$. If the thickness of the slab is $d = 2$ m, determine

- the equation of trajectory of the ray inside the slab
- the coordinates of point where the ray exits from the slab.
- the angle the ray makes with normal at the point of exit.

15. The figure shows the path of a ray passing through an equilateral prism PQR . It is incident on face PR at an angle slightly greater than the critical angle for total internal reflection. If the angle α shown in the figure is 30° , calculate the refractive index of the material of the prism.



16. The cross-section of a glass prism has the form of an isosceles triangle. One of the refracting faces is silvered. A ray of light falling normally on the other refracting face, reflecting twice emerges through the base of the prism perpendicular to it. Find the angles of the prism.